

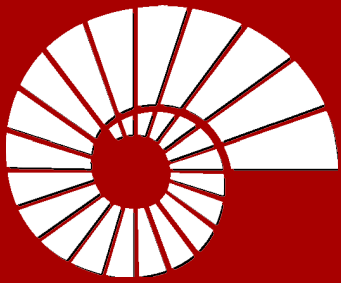
KOÇ UNIVERSITY

Mechanical Engineering



Wheels in Motion:
Empowering Mobility with
Motorized
Wheelchairs

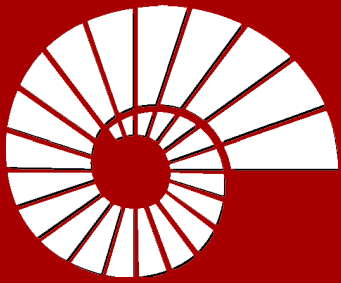
Yiğiter TUNCER
Yusuf CEBECİ
Taha Yusuf YÜCETÜRK



Project Goals

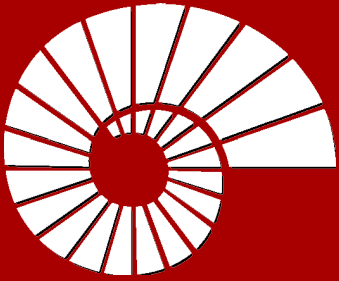
- Custom Retrofitting (Under 20 seconds for attachment)
- Safety and Accessibility
- User-Focused Design
- Cost-Effective Manufacturing (Reusing Hoverboard and Wheelchair)
- Sustainable Design
- Economic Impact (Creating a kit)
- Thorough Testing and User Feedback



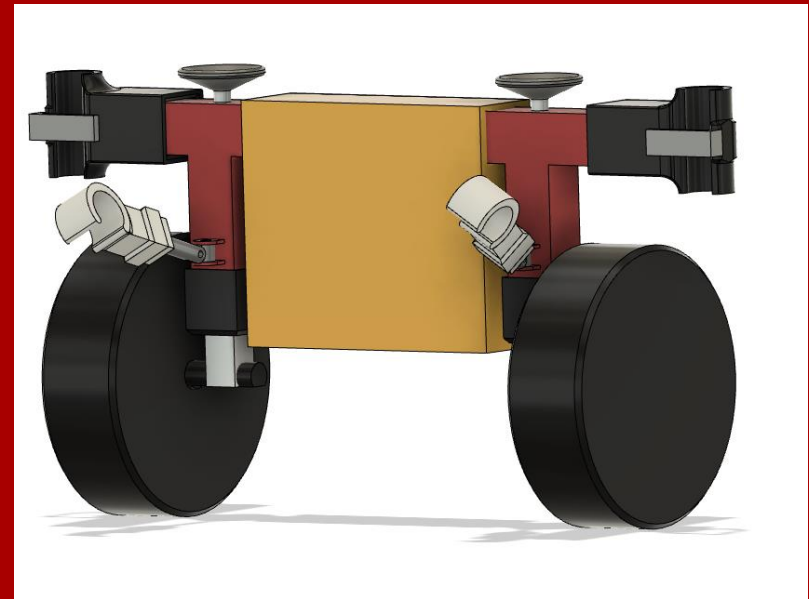


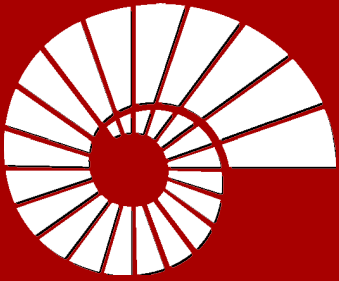
Design Demonstration



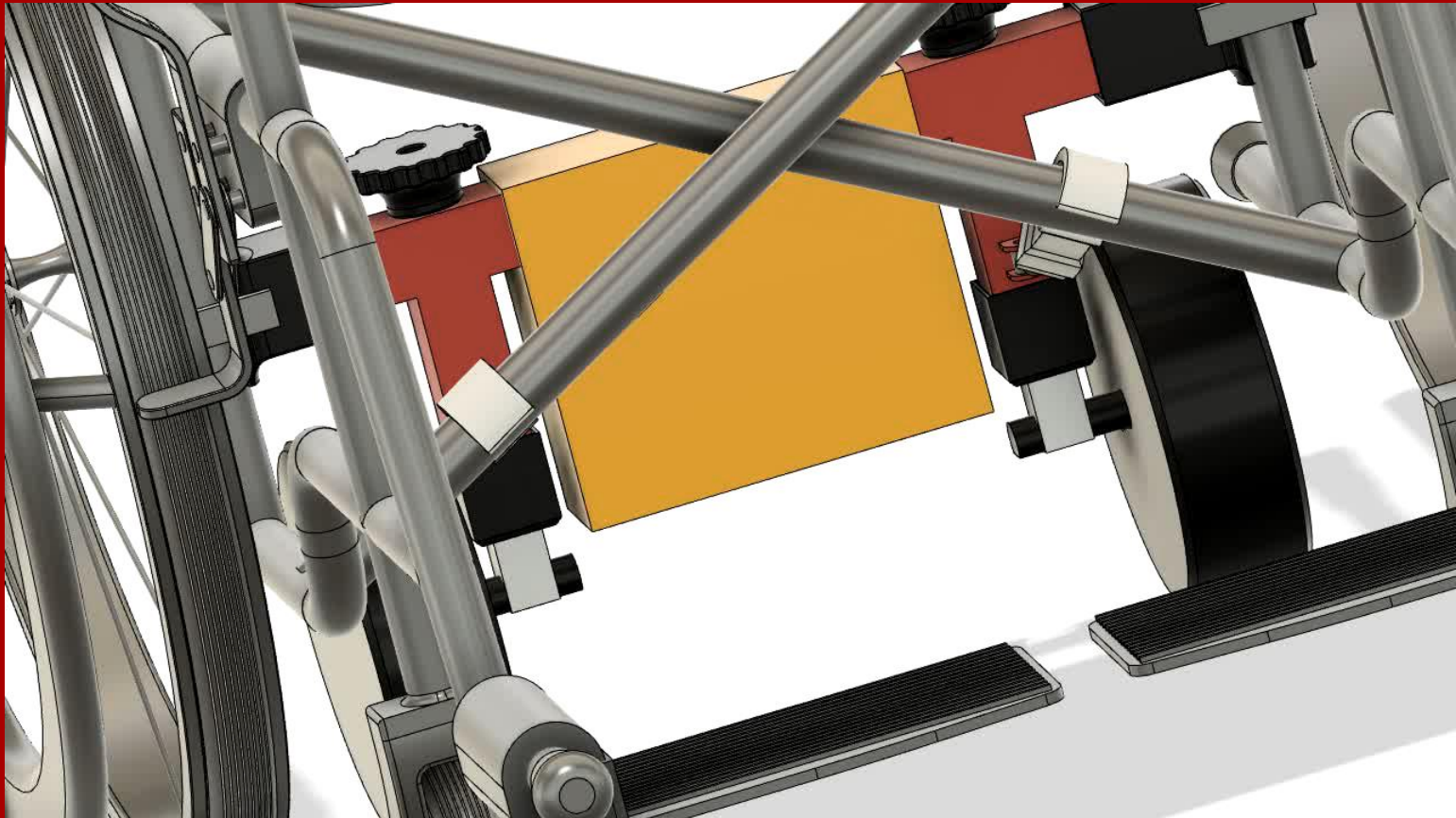


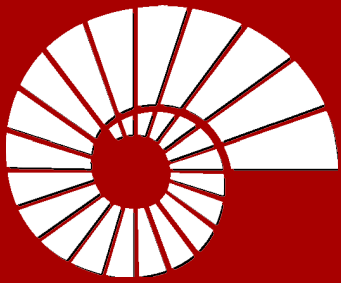
CAD Drawings





Exploded View

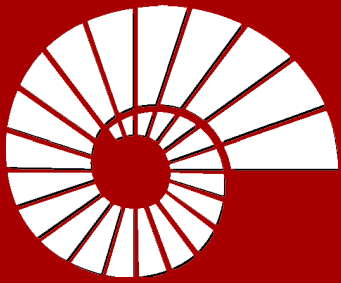




Designed Elements

Hoverboard motor detachment and installation



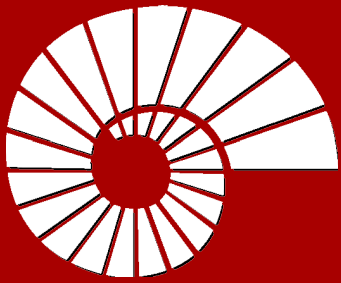


Designed Elements

Main Mounts

3D printed parts assembled together
with clips

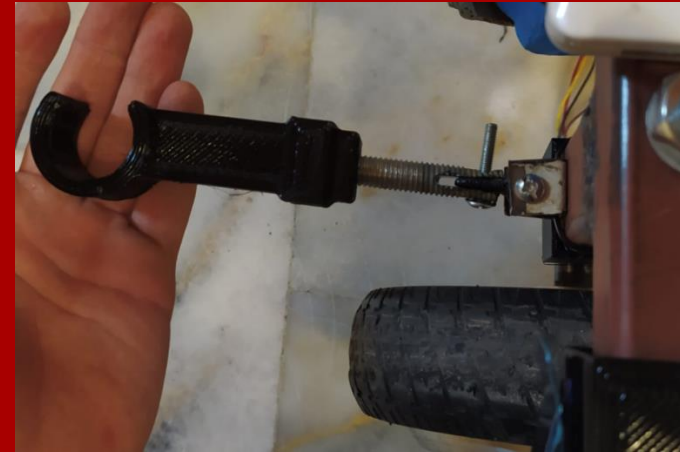
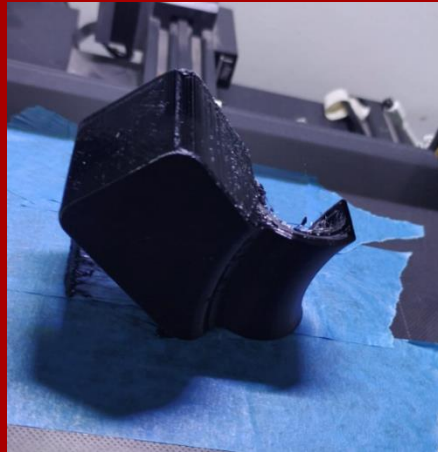


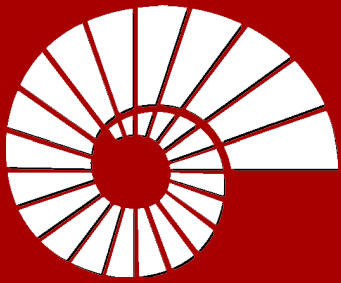


Designed Elements

3D printed parts for retrofitting

- Joystick casing
- Secondary mount



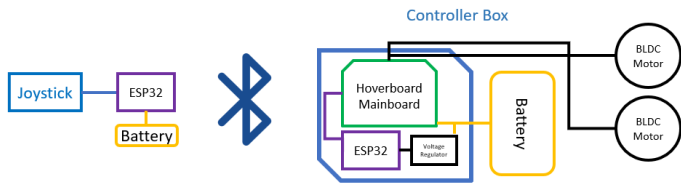


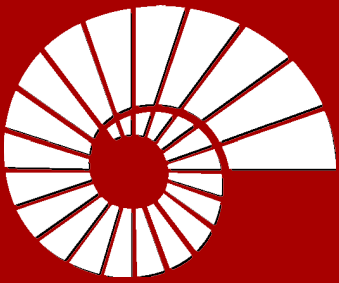
Designed Elements

Joystick Connection

- Used two ESP32 to connect the joystick via bluetooth

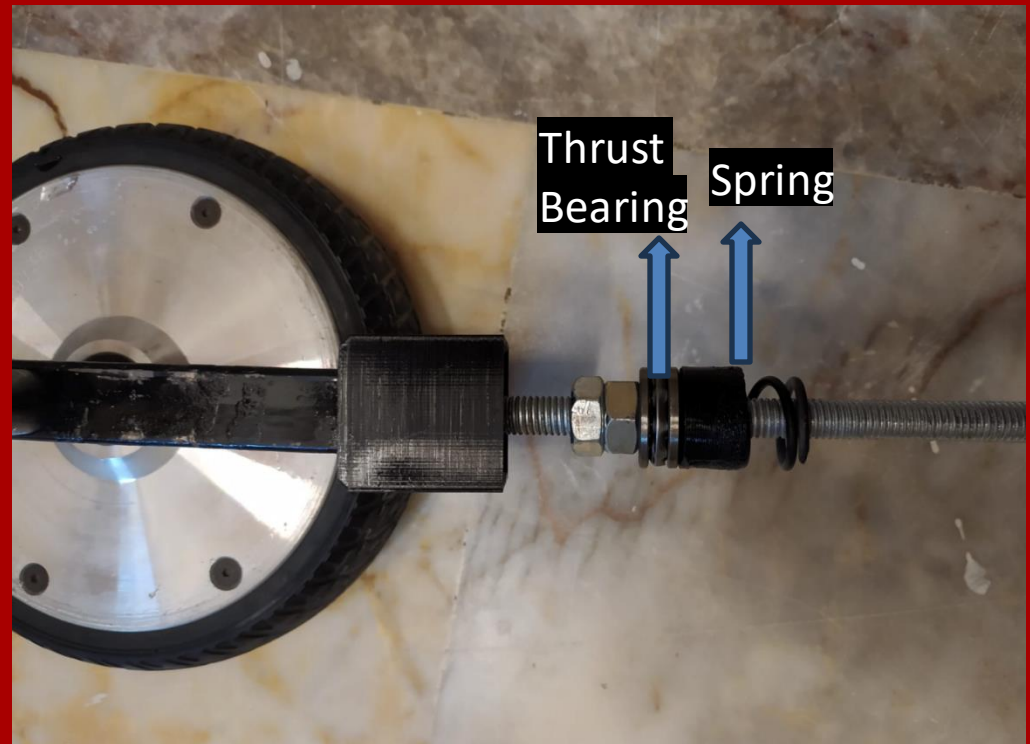
Hoverboard Wheelchair Kit System

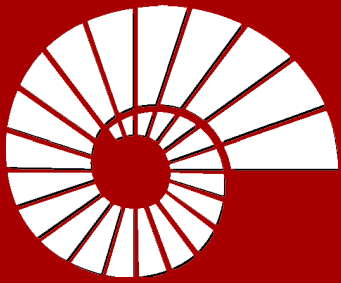




Designed Elements

Spring System on Wheels



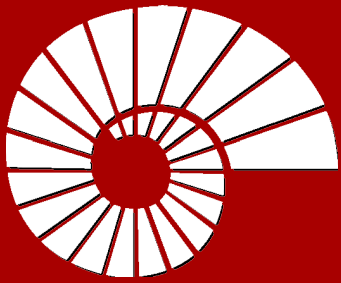


Designed Elements

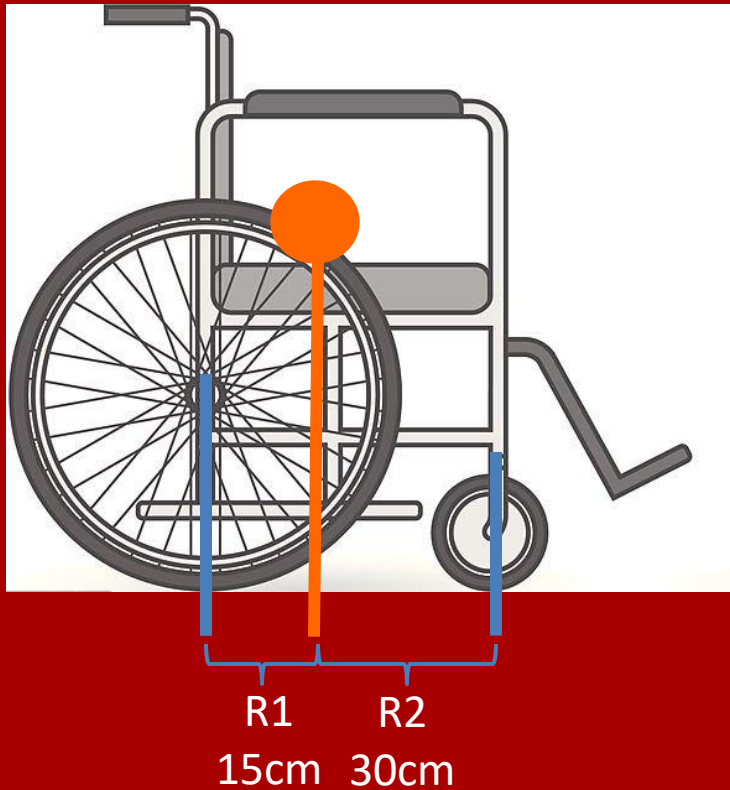
Water Proof Container for Electronics

- Classification: IP65
- Complete protection of dust
- Proof against pressured water (Durable to rain etc.)





Technical Analysis



$$M_{total} * r_1 / r_2 = M_{front} = 103 * 0.33 = 34.33$$

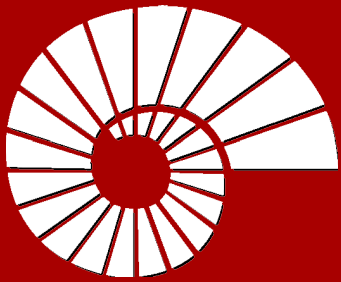
$$M_{total} * r_2 / r_1 = M_{rear} = 103 * 0.67 = 68.67$$

Sitting Person: 73 kg

Wheelchair: 17.30 kg

Hoverboard motors: 9 kg

Total = 103 kg



Technical Analysis

Rear Wheel Contact Area: $6.5 * 1 \text{ cm}^2$,
Hoverboard Motor Contact Area $2.5 * 2 \text{ cm}^2$
 $68.67 * 6.5 / 11.5 = 38\text{-}39 \text{ kg}$ for 2 rear wheels
 $68.67 * 5 / 11.5 = 30 \text{ kg}$ for 2 hoverboard wheels

For the total friction force effecting:

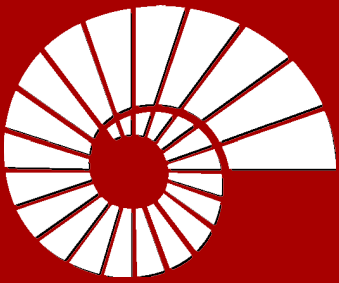
$F_{fr}(\text{force of friction rolling}) = m * g * \mu = 103 * 9.81 * 0.02 = 20 \text{ N} = 2 \text{ kg}$ force forward
The acceleration of the wheelchair is making the most resistance and load in the design. Planned acceleration of our wheelchair: 1.7 m/s .

$$F = m * a$$

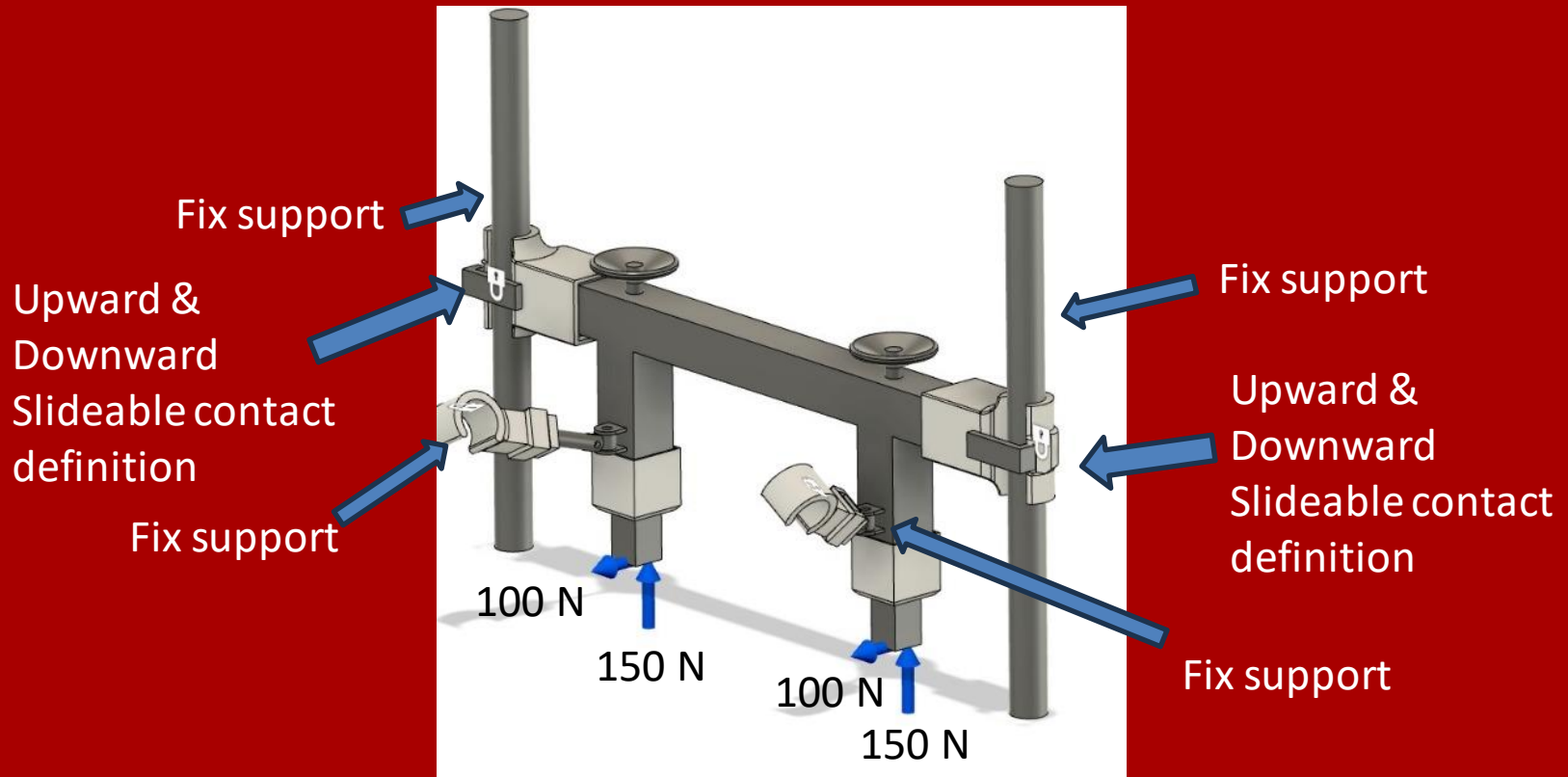
$$F = 103 * 1.7 = 175.1 \text{ N}$$

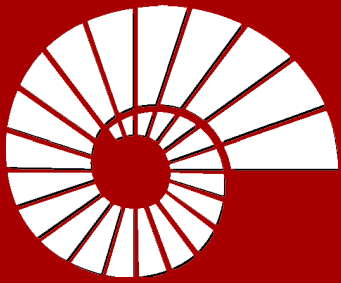
$$F_{net} = F + F_{fr} = 17.5 \text{ kg} + 2 \text{ kg} \approx 20 \text{ kg}$$





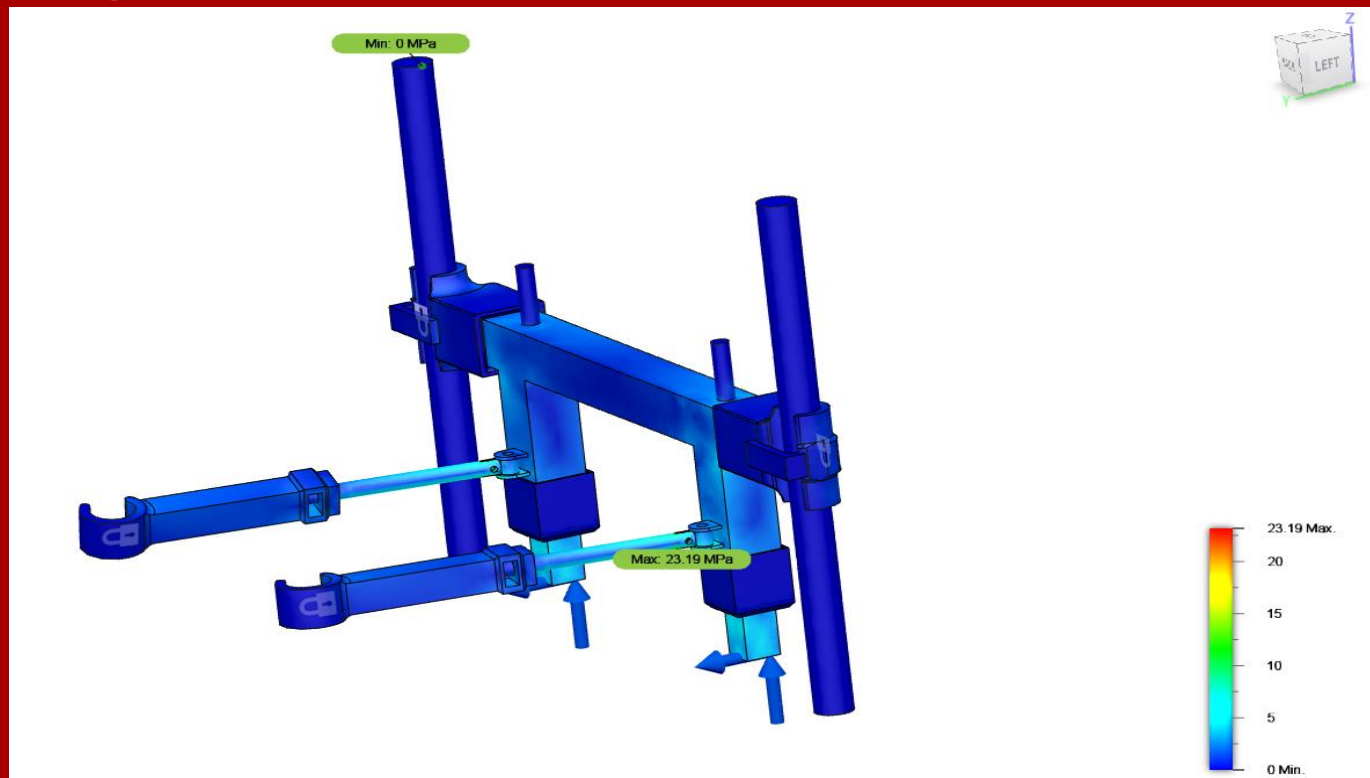
Technical Analysis

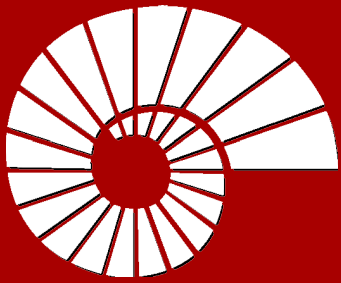




Technical Analysis

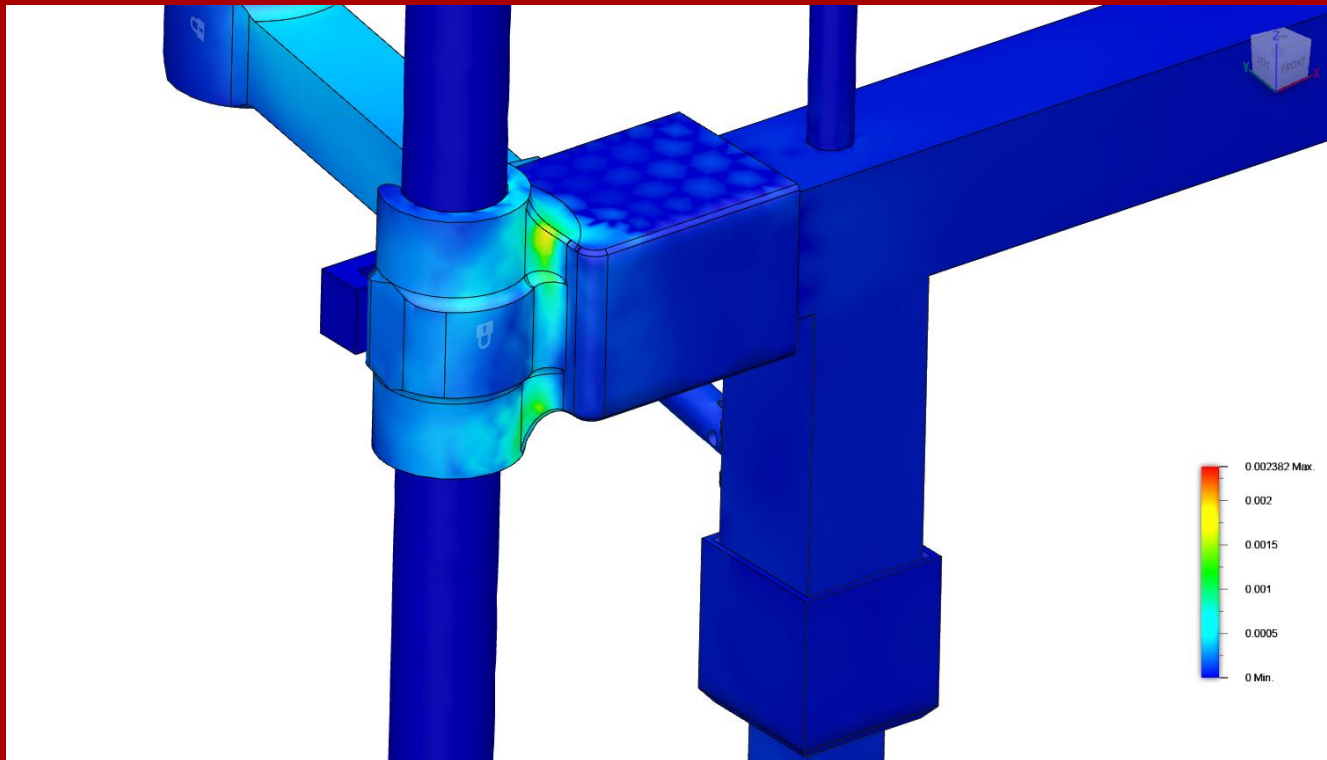
Stress Analysis

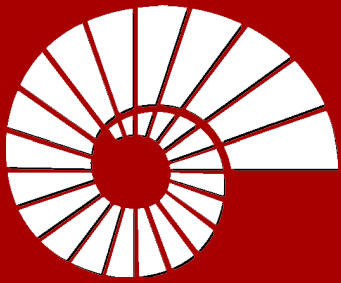




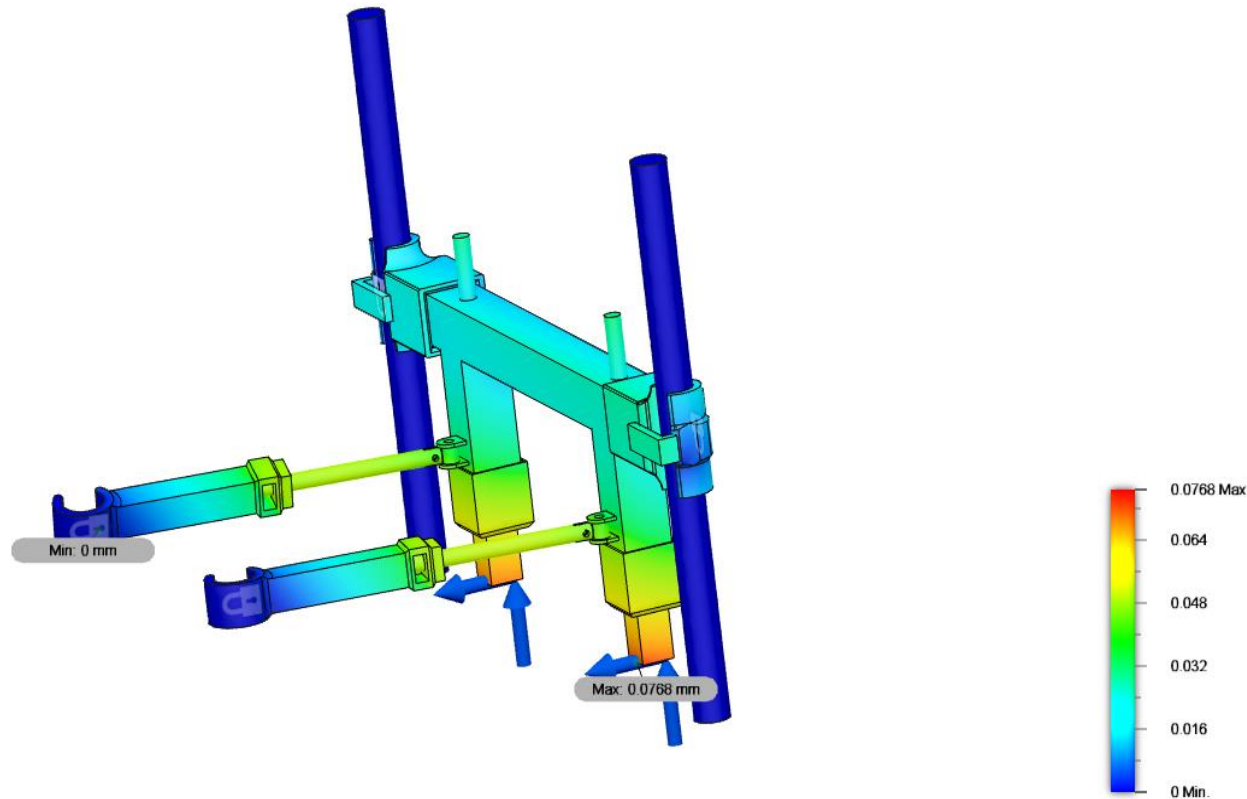
Technical Analysis

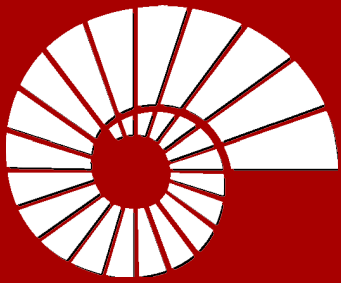
Strain Analysis



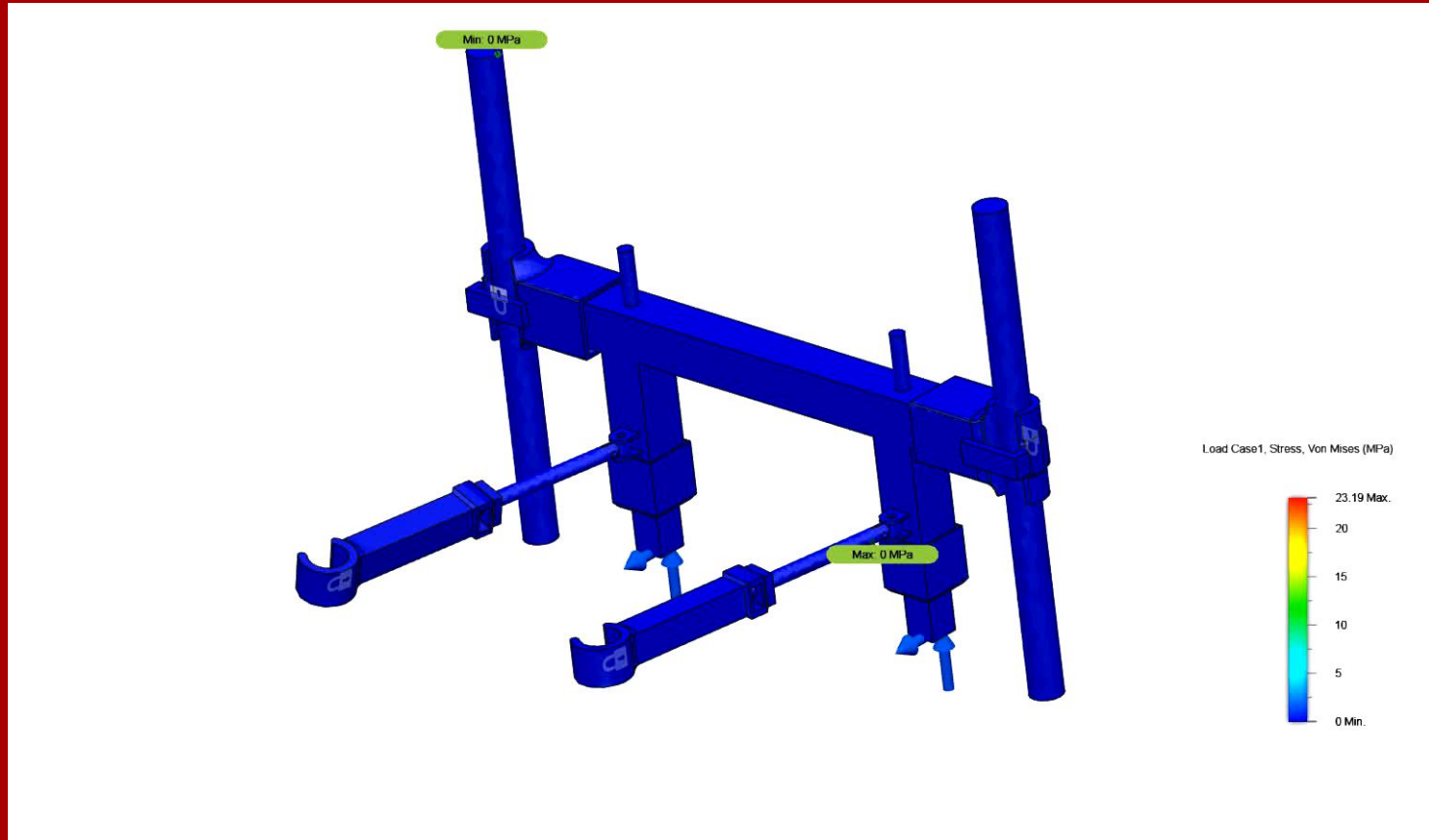


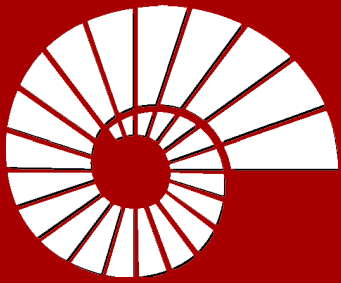
Displacement Analysis





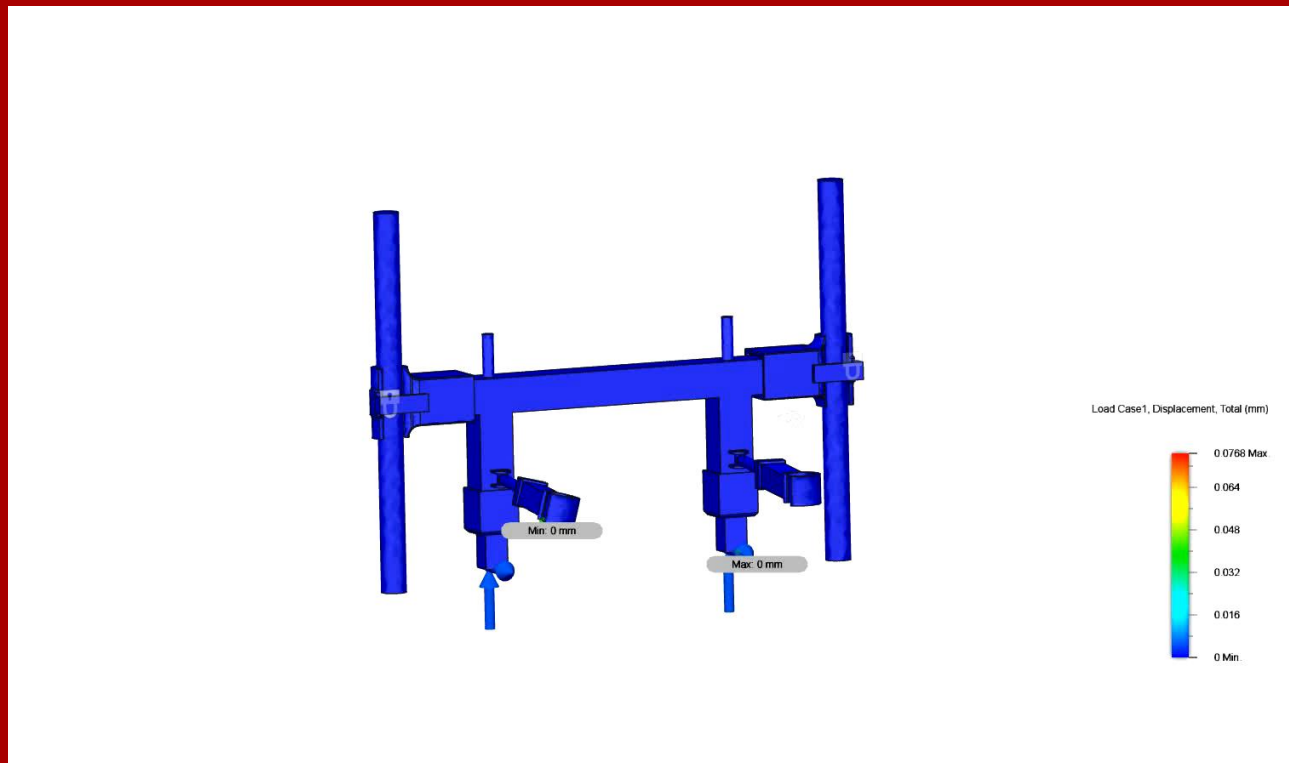
Stress Result Animation

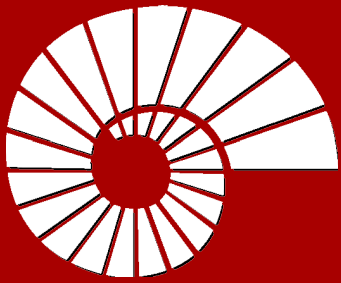




Major Results

Adjusted FEM Results





Conclusion and Future of the Design

Project Goals Achieved:

- Custom retrofitting for electrical design
- Emphasis on safety assessments
- User-focused designs implemented
- Utilization of renewable materials in production
- Consideration of economic impact

Prototype Highlights:

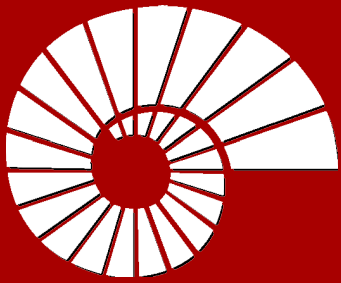
- Utilization of repurposed hoverboard motors for both manual and electric modes in under 20 seconds
- Adaptability to most manual wheelchair models via adjustable parts
- Waterproof electronics container with IP65 classification for durability

Analysis and Assurance:

- Comprehensive analysis addressing center of gravity, mass distribution, and forces affecting design
- Stress and strain analysis within acceptable limits
- Minimal displacement (0.07mm) ensuring structural integrity

Future Prospects:

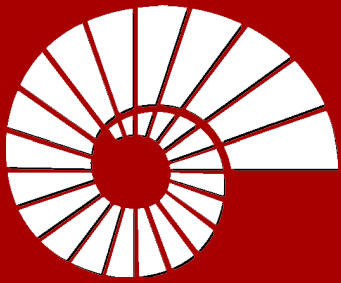
- Refinement and improvement of current design
- Aim for wider accessibility and inclusivity
- Potential collaborations for enhanced iterations
- Continued focus on sustainability and cost-effectiveness



Acknowledgement

- Emir Bahadır Ünsal (TA)
- Ease AliAbbasi (TA)
- Enes Ulaş Dinçer (TA)





Thank you!

References

- [1] Requejo, P. S., Furumasu, J., & Mulroy, S. J. (2015). Evidence-based strategies for preserving mobility for elderly and aging manual wheelchair users. *Topics in Geriatric Rehabilitation*, 31(1), 26–41. <https://doi.org/10.1097/tgr.0000000000000042>
- [2] Zimmerman, M. (n.d.). Technology and handicapped people May 1982 - Princeton University. <https://www.princeton.edu/~ota/disk3/1982/8229/822901.PDF>
- [3] Flemmer, C. L. (2022). *Improving the built environment for manual wheelchair users: A review...* IOP Science. <https://iopscience.iop.org/article/10.1088/1755-1315/1101/3/032031>

